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Summary

I design, build and operate an MLOps platform on on-prem Kubernetes, alone. Over six months at HUIINNO I codified the entire path — ingestion → lakehouse → transformation → lineage → distributed training → experiment / model registry — in **Pulumi (Python IaC)**, as the **sole owner of cluster construction** on a two-person platform team. I remove human bottlenecks rather than add tools, and decide by measurement. Earlier: iOS → backend → frontend → cloud data pipelines, then led a data engineering team.

Key Results (measured 2026-09)

Bare-metal Kubernetes reproducible in code — ingestion to model registry modularized into 12 Pulumi stacks; infrastructure changes tracked in PRs, new environments rebuilt by composing stacks. (Authored 93% of stack-code changes — 6 above 96% — and 92% of 1,204 commits; 678 merged PRs.)

10 nodes / 235 running pods / 23 internal services in production. SSO across all 23 with Dex (GitHub OIDC) + oauth2-proxy, the same identity carried into Kubernetes RBAC.

Removed the single-person dependency in dataset preparation — “ask one data scientist and wait” became scheduled Airflow 3.x + dbt + Apache Iceberg runs with lineage in DataHub; researchers now query directly. (Authored 83% of the ETL repo’s 1,599 commits, 93% of the shared framework, 100% each of CAST, MIMIC-IV and operational pipelines.)

Training-data delivery 25× faster — network and storage redesign took iperf3 throughput from 93.7 Mbps to 2.5 Gbps; an 8 TB transfer from 8.4 days to 7.8 hours.

Reversed a GPU purchase decision with measurement — 10 days of telemetry showed 5.3% utilization and zero queued jobs; the bottleneck was 1,112 GPU-hours held idle by dormant sessions (91% of held time); redirected from purchase to reclamation.

AI coding agents run continuously to lift one person’s throughput to team scale — backed by 719 SDK unit tests, CI gates, 23 incident write-ups, 30 PRDs.

Earlier: Led the AWS EMR / PySpark art-market ETL pipeline (Eazel, team lead).

What I Do Now

Data Engineer, HUIINNO (Dec 2025 -). I operate every layer of the MLOps platform on bare-metal Kubernetes and am attaching a reproducibility contract (version, content hash, code commit, environment lock and seed stamped onto a dataset at run time) plus MLflow tracking and a model registry, so research code can move onto KubeRay. A quantified survey found the root cause — results having no coordinates — and I am establishing the split where researchers own domain ETL and the platform owns execution, lineage, verification and retries.

Stack

Platform: Kubernetes (kubeadm) · Pulumi · Cilium · SeaweedFS · Dex/OIDC · Prometheus/Grafana

Data: Airflow 3.x · dbt · Apache Iceberg · DataHub · PostgreSQL (pg_duckdb) · ClickHouse

ML: Ray/KubeRay · MLflow · PyTorch · DVC

Languages & other: Python · SQL · Swift · Shell · AWS/GCP · LangChain

Certifications & Languages

Certifications: Engineer Information Processing (Nov 2017) · Craftsman Information Processing (Jul 2009) · HRD Korea

Languages: Korean (native) · English AL (OPIc) · Japanese (conversational)

Full experience and project details are in a separate document ([resume_english.pdf](#)).